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CSA1607

Cloud computing and Big Data Analytics CO2 - AT2.

Q1) Data preprocessing in Retail Analytics

A. Handling Missing values.

i) Missing values.

- Age → C2
- Monthly.spend → C3
- purchase_frequency → C5.

ii) Imputation Techniques

- Median Imputation
- Age = Median of (23, 31, 105, 27, 40, 36, 42) = 36.
- Monthly_spend = Median of (20, 35, 500, 25, 45, 38, 300) = 38
- purchase_frequency = Median of (5, 8, 10, 50, 12, 9, 45) = 10
- K-NN Imputation.
- Missing values are filled using values of similar customers

iii) Justification

- Median handles outlier better.
- K-NN gives accurate estimates based on similar records.

B. Outlier Detection

- outliers :

- Monthly_spend = 500
- Treatment: Cap Age to 60 and spend to 300
- Reason: Prevents extreme values from affecting analysis

c. categorical standardization

| original | standardized |
|---|---|
| M | Male |
| Male | Male |
| Male | Male |
| F | Female |
| FEMALE | Female. |
| Female | Female. |

Result:- consistent gender values improve data quality

2) covariance & Data Reduction:

A. Covariance Analysis:-

- Income and Purchase_Value increases together.
- covariance is Positive (+ve)

Interpretation: Higher. income customers generally spend more

Use of covariance:-

- Identifies relationships between variables.
- Help detect redundant features.

B. Feature Selection:-

Redundant features

- Income.
- Avg_Balance
- EMI (highly related)

Selected features:-

- Age
- Income
- purchase_Value.
- credit_score
- orders.

1000 1.77.

Reason:- These features contain enough information for prediction.

C. Data Transformation

- Standardize all numeric columns:

Why?

- Bring all features to the same scale.
- Improves model accuracy and speed

contains al

3) Data Transformation & Discretization.

A. Equal-Width Bining.

Study Hours

- Min=2, Max=11
- Bin Width= (11-2)/3 =3

Bins:-

- 2-5 → low
- 5-8 → Medium
- 8-11 → High

Test Score

- Min=40, Max=100
- Bin width = (100-40)/3 = 20

Bins :

- 40-60
- 60-80
- 80-100

B. Equal-frequency Bining.

Each bin contains about 3-4 records

[illegible]

[illegible]

Study Hours

- 1-4 → Low
- 5-7 → Medium
- 8-11 → High

Test Score

- 40-59 → Poor
- 60-79 → Average
- 80-100 → Excellent

4. Normalization

Given Data.

200, 300, 400, 600, 1000

A. Min-Max Normalization [0,1]

formula:

| Original | Normalized |
|---|---|
| 200 | 0.000 |
| 300 | 0.125 |
| 400 | 0.250 |
| 600 | 0.500 |
| 1000 | 1.000 |

B. Z-score Normalization

Mean = 500

Standard Deviation ≈ 282.84

| Value | Z-Score |
|---|---|
| 200 | -1.06 |
| 300 | -0.71 |
| 400 | -0.35 |
| 600 | 0.35 |
| 1000 | 1.77. |

Conclusion:-

- Min-Max scales data between 0 and 1
- Z-score centers data around 0 with standard deviation 1

- 25,23,28

5) Online Shopping Behaviour Analysis

A) Equal-frequency Binning

Bin1 : 15, 30, 45, 60

Bin2 120, 300, 600

Bin3 900, 1800, 3600

Analysis

- Short visits are grouped together.
- Long visits form separate groups, making customer behaviour easier to analyze.

B) Z-score Normalization

- Mean = 747 sec
- Standard deviation is calculated.
- Formula :

Effect

- large values (1800, 3600) get high positive z-scores
- Makes comparison easier.

c) Min-Max Normalization

Formula:

| Duration | Normalized. |
|---|---|
| 15 | 0.000 |
| 30 | 0.004 |
| 45 | 0.008 |
| 60 | 0.013 |
| 120 | 0.029 |
| 300 | 0.080 |
| 600 | 0.163 |
| 900 | 0.247 |
| 1800 | 0.498 |
| 3600 | 1.000 |

Comparison

- Min-Max: Values lie between 0 and 1, simple for clustering
- Z-score: Better when data contains extreme values and different scales.